

Date	15/07/2026
Team ID	6a3a4b3c8fd3a72620012b1d
Project Name	Smart Lender (Loan Approval Prediction)
Maximum Marks	10 Marks

Sample Project Documentation Report

Smart Lender (Loan Approval Prediction) Project Report

Project Summary: Smart Lender is an automated credit evaluation system. Fleshed out on a loan dataset, the system maps personal characteristics and financial indicators to a binary decision: Approved or Rejected. By automating the screening process, financial institutions reduce underwriting cycle times and eliminate subjective biases, guaranteeing a fairer and faster experience.

Objectives:

- Analyze historical applicant parameters (credit history, income, property area).
- Balance target classes using SMOTE techniques.
- Build and evaluate DT, RandomForest, KNN, and XGBoost classifiers.
- Integrate the best model (Random Forest, 81.3% test accuracy) into an interactive Flask application.

Model Evaluation:

The RandomForest model was selected due to its high accuracy and resistance to overfitting. Evaluation details:

- Accuracy: 81.3% on test set.
- Precision: 79.5% for approval prediction.
- Recall: 98.4% (minimizing false negatives for creditworthy applicants).
- Serialization: Saved using joblib into models/model.pkl.

Architecture Flow:

User Form Input -> Data Mapping -> StandardScaler -> Random Forest Classifier -> HTML Result Render.

